Physics for Scientists and Engineers Storyline

Tenth Edition

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Chapter 22

Electric Fields

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Properties of Electric Charges (1 of 2)



a

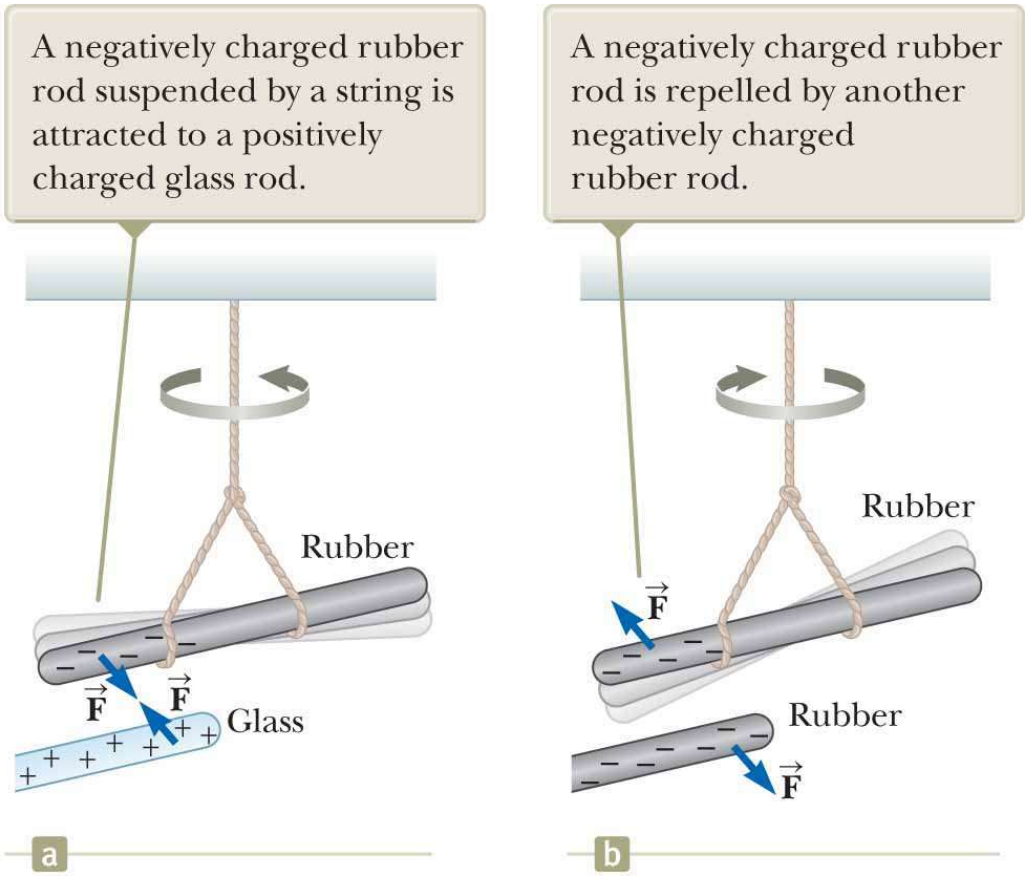


b

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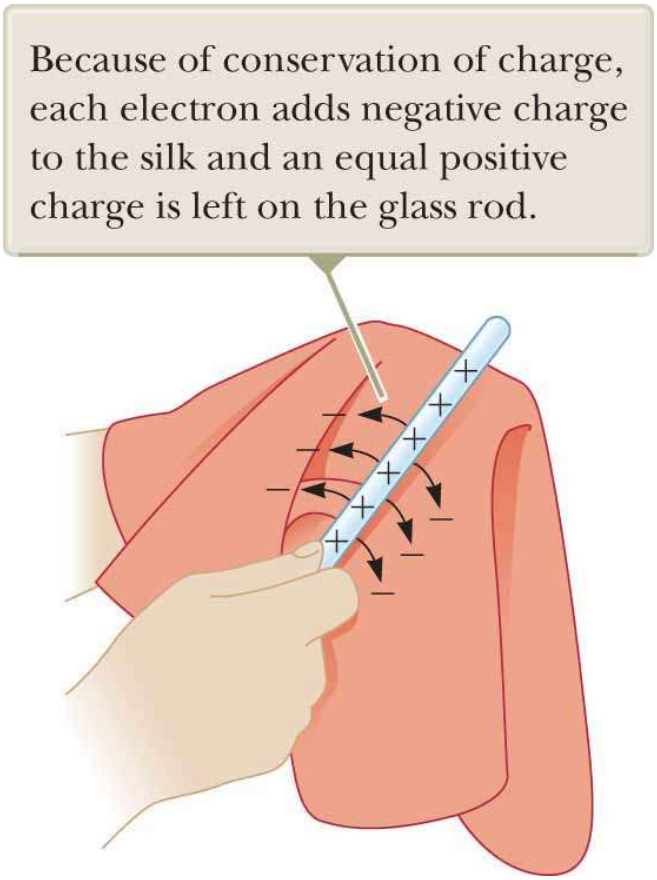
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Properties of Electric Charges (2 of 2)



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Conservation of Charge



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Charge is Quantized



Robert A. Millikan

$$q = \pm Ne$$

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Quick Quiz 22.1 (1 of 2)

Three objects are brought close to each other, two at a time. When objects A and B are brought together, they repel. When objects B and C are brought together, they also repel. Which of the following are true?

(a) Objects A and C possess charges of the same sign.

(b) Objects A and C possess charges of opposite sign.

(c) All three objects possess charges of the same sign.

(d) One object is neutral.

(e) Additional experiments must be performed to determine the signs of the charges.

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Quick Quiz 22.1 (2 of 2)

Three objects are brought close to each other, two at a time. When objects A and B are brought together, they repel. When objects B and C are brought together, they also repel. Which of the following are true?

(a) Objects A and C possess charges of the same sign.

(b) Objects A and C possess charges of opposite sign.

(c) All three objects possess charges of the same sign.

(d) One object is neutral.

(e) Additional experiments must be performed to determine the signs of the charges.

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Insulators and Conductors

Electrical conductors are materials in which some of the electrons are free electrons that are not bound to atoms and can move relatively freely through the material; electrical insulators are materials in which all electrons are bound to atoms and cannot move freely through the material.

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Semiconductors



silicon

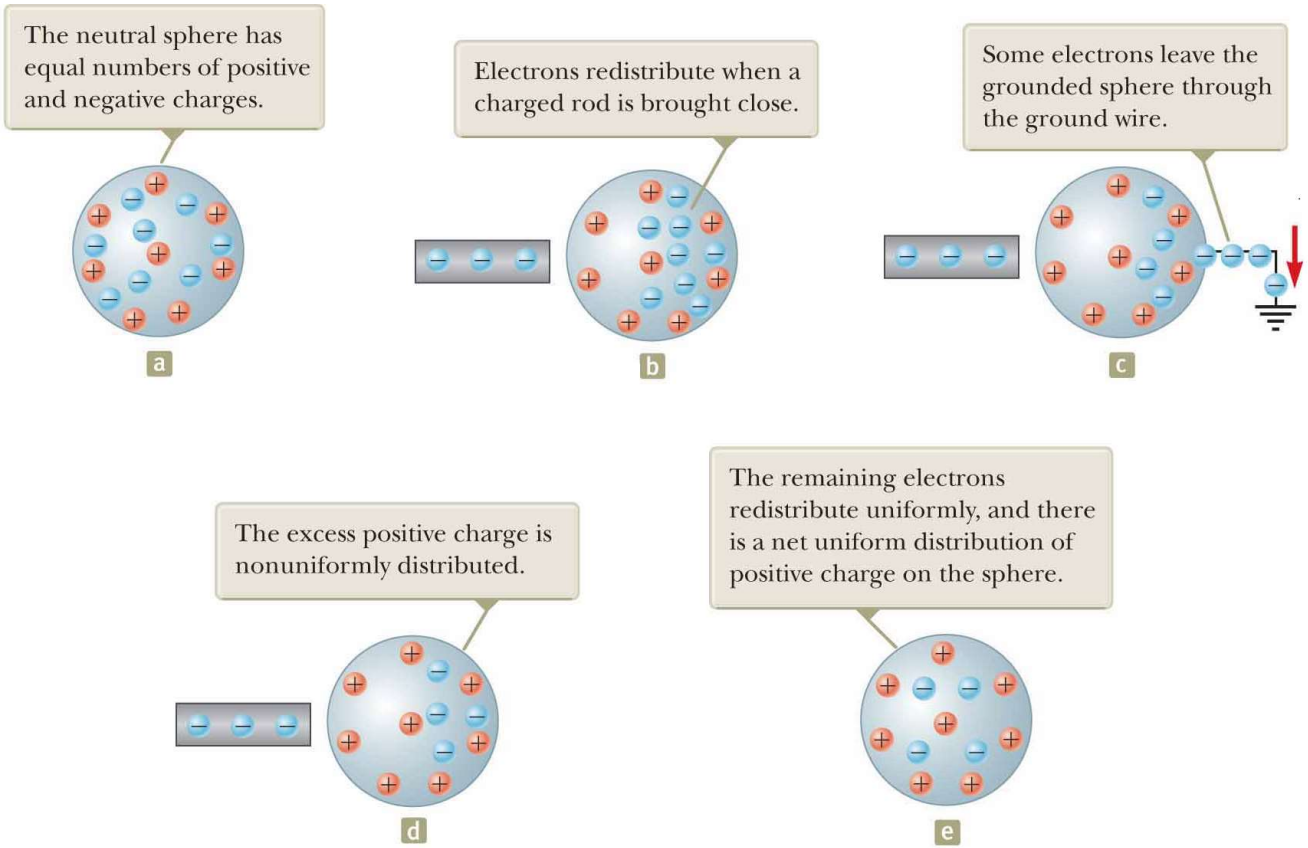


polycrystalline germanium

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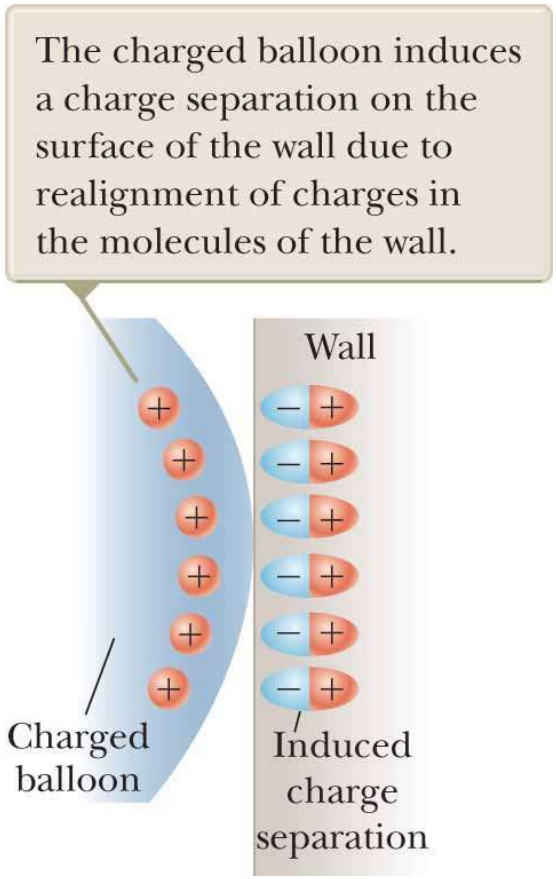
Charging Objects by Induction (1 of 2)



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Charging Objects by Induction (2 of 2)



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Quick Quiz 22.2 (1 of 2)

Three objects are brought close to one another, two at a time. When objects A and B are brought together, they attract. When objects B and C are brought together, they repel. Which of the following are necessarily true?

(a) Objects A and C possess charges of the same sign.

(b) Objects A and C possess charges of opposite sign.

(c) All three objects possess charges of the same sign.

(d) One object is neutral.

(e) Additional experiments must be performed to determine information about the charges on the objects.

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Quick Quiz 22.2 (2 of 2)

Three objects are brought close to one another, two at a time. When objects A and B are brought together, they attract. When objects B and C are brought together, they repel. Which of the following are necessarily true?

(a) Objects A and C possess charges of the same sign.

(b) Objects A and C possess charges of opposite sign.

(c) All three objects possess charges of the same sign.

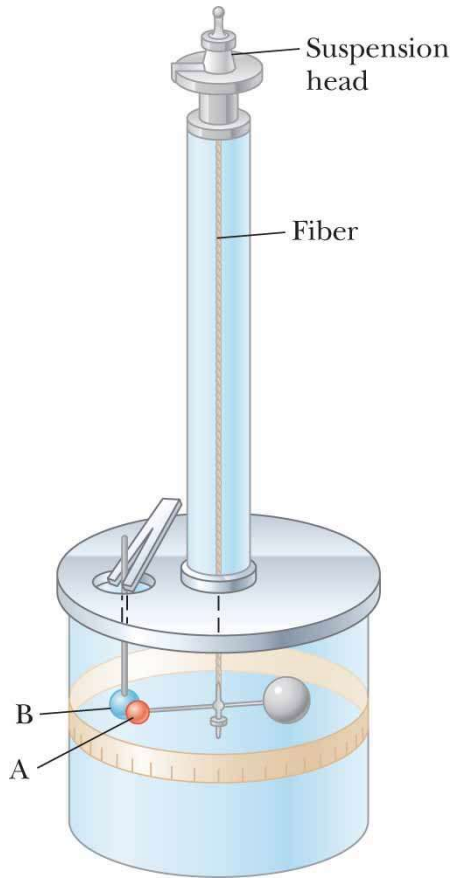
(d) One object is neutral.

(e) Additional experiments must be performed to determine information about the charges on the objects.

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Electric Forces between Charged Objects



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Coulomb's Law

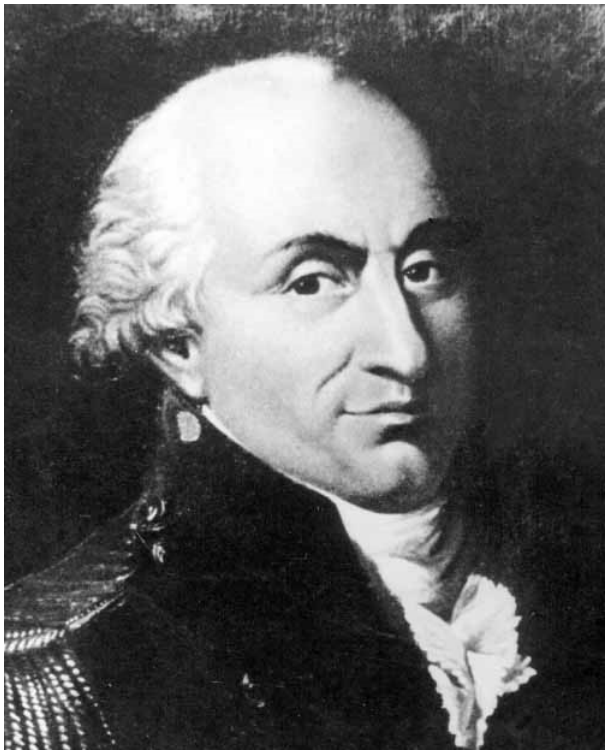
$$F_e = k_e \frac{|q_1||q_2|}{r^2}$$

$$k_e = 8.9876 \times 10^9 \ \mathrm{N \cdot m^2/C^2}$$

$$k_e = \frac{1}{4\pi\varepsilon_0}$$

$$\varepsilon_0 = 8.8542 \times 10^{-12} \ \mathrm{C^2 \cdot N^{-1} \cdot m^{-2}}$$

$$e = 1.60218 \times 10^{-19} \ \mathrm{C}$$

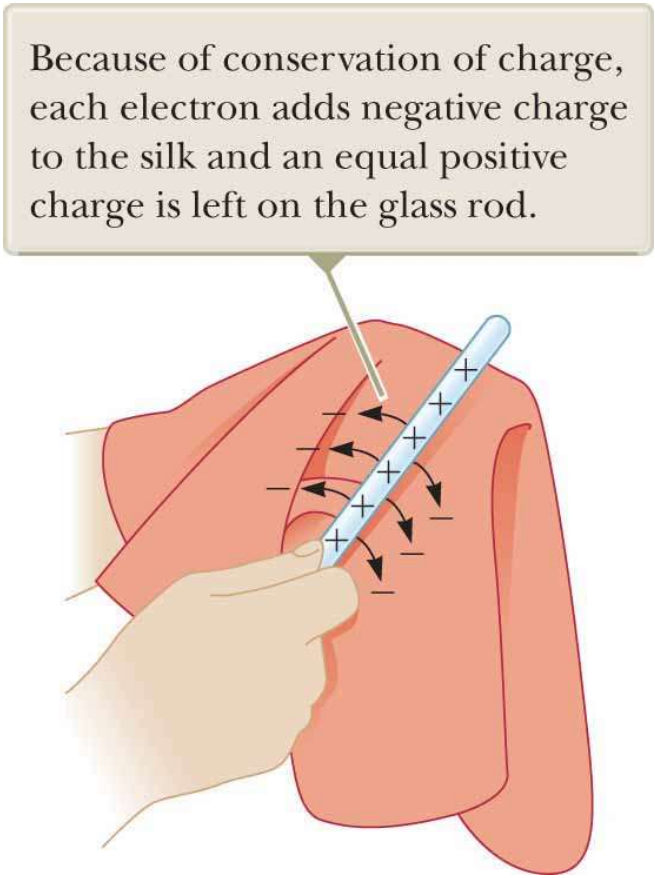


Charles Coulomb

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Coulomb of Charge



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Charges and Masses of Electrons, Protons, and Neutrons

Table 22.1 Charge and Mass of the Electron, Proton, and Neutron

Particle	Charge (C)	Mass (kg)
Electron (e)	$-1.602\,176\,6 \times 10^{-19}$	$9.109\,4 \times 10^{-31}$
Proton (p)	$+1.602\,176\,6 \times 10^{-19}$	$1.672\,62 \times 10^{-27}$
Neutron (n)	0	$1.674\,93 \times 10^{-27}$

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Example 22.1: The Hydrogen Atom (1 of 2)

The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately 5.3×10^{-11} m. Find the magnitudes of the electric force and the gravitational force between the two particles.

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Example 22.1: The Hydrogen Atom (2 of 2)

$$F_e = k_e \frac{|e||-e|}{r^2} = \left(8.988\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\frac{\left(1.60\times10^{-19}\ \text{C}\right)^2}{\left(5.3\times10^{-11}\ \text{m}\right)^2}$$

$$= \boxed{8.2\times10^{-8}\ \text{N}}$$

$$F_g = G\frac{m_e m_p}{r^2}$$

$$= \left(6.674\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\frac{\left(9.11\times10^{-31}\ \text{kg}\right)\left(1.67\times10^{-27}\ \text{kg}\right)}{\left(5.3\times10^{-11}\ \text{m}\right)^2}$$

$$= 3.6\times10^{-47}\ \text{N}$$

$$\frac{F_e}{F_g} = \frac{8.2\times10^{-8}\ \text{N}}{3.6\times10^{-47}\ \text{N}} \approx 2\times10^{39}$$

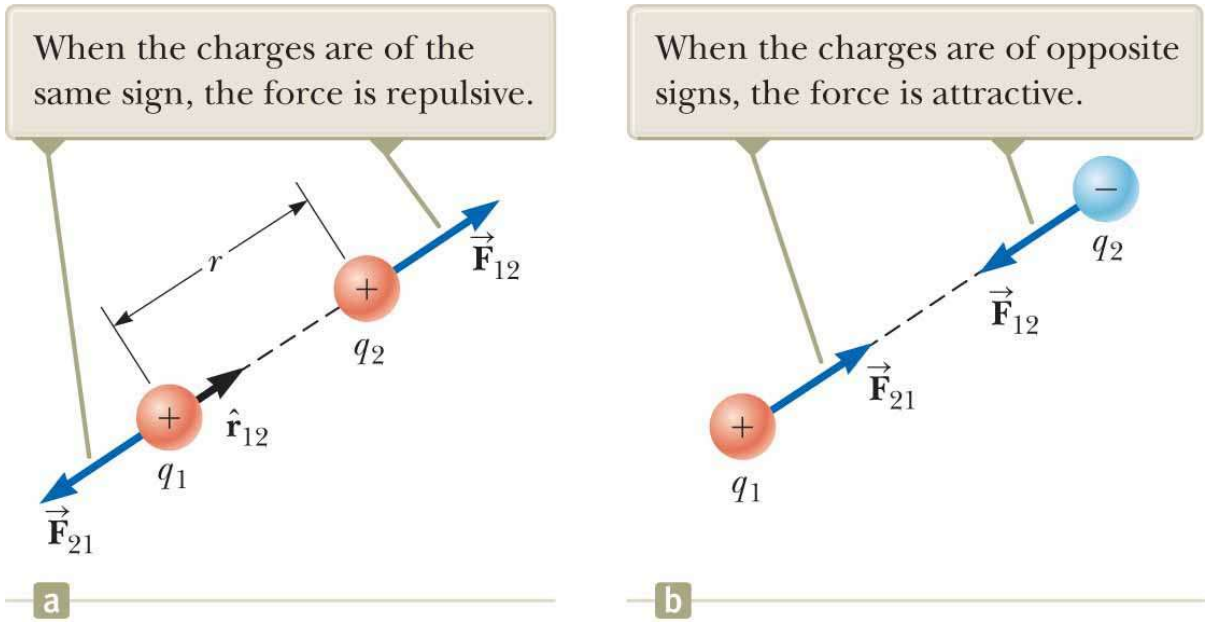
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Coulomb's Law in Vector Form

$$\vec{\mathbf{F}}_{12} = k_e \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{12}$$

$$\sum \vec{\mathbf{F}}_1 = \vec{\mathbf{F}}_{21} + \vec{\mathbf{F}}_{31} + \vec{\mathbf{F}}_{41}$$



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Quick Quiz 22.3 (1 of 2)

Object A has a charge of +2 μC, and object B has a charge of +6 μ C. Which statement is true about the electric forces on the objects?

(a) $\vec{\mathbf{F}}_{AB} = -3\vec{\mathbf{F}}_{BA}$

(b) $\vec{\mathbf{F}}_{AB} = -\vec{\mathbf{F}}_{BA}$

(c) $3\vec{\mathbf{F}}_{AB} = -\vec{\mathbf{F}}_{BA}$

(d) $\vec{\mathbf{F}}_{AB} = 3\vec{\mathbf{F}}_{BA}$

(e) $\vec{\mathbf{F}}_{AB} = \vec{\mathbf{F}}_{BA}$

(f) $3\vec{\mathbf{F}}_{AB} = \vec{\mathbf{F}}_{BA}$

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Quick Quiz 22.3 (2 of 2)

Object A has a charge of +2 μC, and object B has a charge of +6 μ C. Which statement is true about the electric forces on the objects?

(a) $\vec{\mathbf{F}}_{AB} = -3\vec{\mathbf{F}}_{BA}$

(b) $\boxed{\vec{\mathbf{F}}_{AB} = -\vec{\mathbf{F}}_{BA}}$

(c) $3\vec{\mathbf{F}}_{AB} = -\vec{\mathbf{F}}_{BA}$

(d) $\vec{\mathbf{F}}_{AB} = 3\vec{\mathbf{F}}_{BA}$

(e) $\vec{\mathbf{F}}_{AB} = \vec{\mathbf{F}}_{BA}$

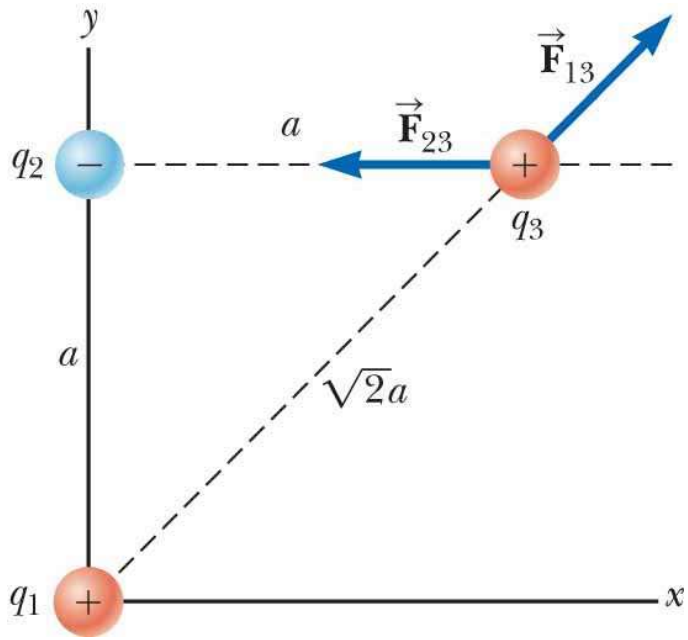
(f) $3\vec{\mathbf{F}}_{AB} = \vec{\mathbf{F}}_{BA}$

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Example 22.2: Find the Resultant Force (1 of 3)

Consider three point charges located at the corners of a right triangle as shown in the figure, where $q_1 = q_3 = 5.00\ \mu C$, $q_2 = -2.00\ \mu C$, and $a = 0.100$ m. Find the resultant force exerted on q_3.



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Example 22.2: Find the Resultant Force (2 of 3)

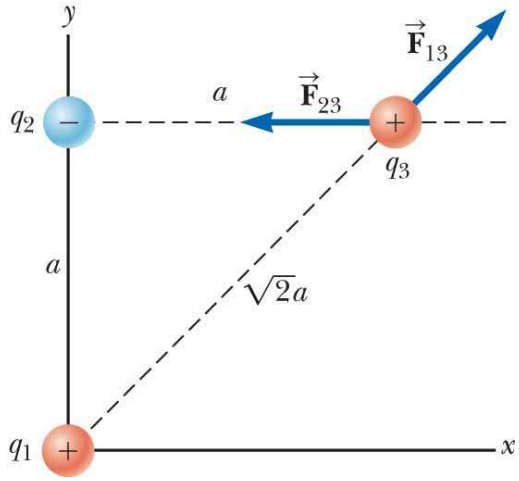
$$F_{23} = k_e \frac{|q_2||q_3|}{a^2} = \left(8.988\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\frac{\left(2.00\times10^{-6}\ \text{C}\right)\left(5.00\times10^{-6}\ \text{C}\right)}{\left(0.100\ \text{m}\right)^2} = 8.99\ \text{N}$$

$$F_{13} = k_e \frac{|q_1||q_3|}{\left(\sqrt{2}a\right)^2} = \left(8.988\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\frac{\left(5.00\times10^{-6}\ \text{C}\right)\left(5.00\times10^{-6}\ \text{C}\right)}{2\left(0.100\ \text{m}\right)^2} = 11.2\ \text{N}$$

$$F_{13x} = \left(11.2\ \text{N}\right)\cos 45.0^\circ = 7.94\ \text{N}$$
$$F_{13y} = \left(11.2\ \text{N}\right)\sin 45.0^\circ = 7.94\ \text{N}$$

$$F_{3x} = F_{13x} + F_{23x} = 7.94\ \text{N} + \left(-8.99\ \text{N}\right) = -1.04\ \text{N}$$
$$F_{3y} = F_{13y} + F_{23y} = 7.94\ \text{N} + 0 = 7.94\ \text{N}$$

$$\vec{\mathbf{F}}_3 = \left(-1.04\hat{\mathbf{i}} + 7.94\hat{\mathbf{j}}\right)\ \text{N}$$

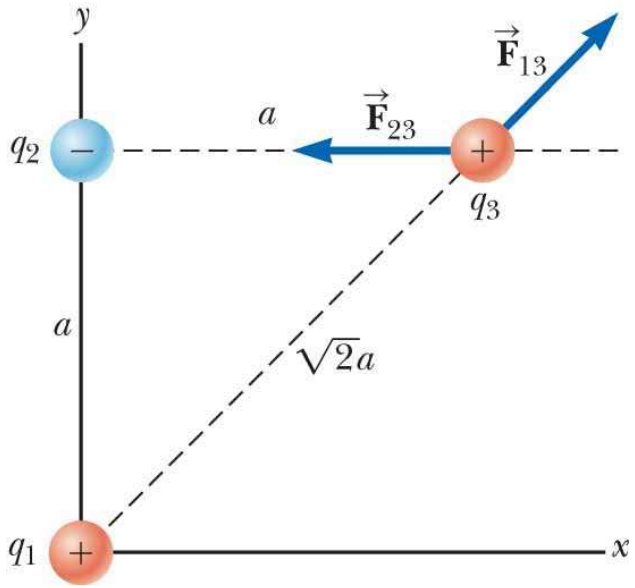


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Example 22.2: Find the Resultant Force (3 of 3)

What if the signs of all three charges were changed to the opposite signs? How would that affect the result for $\vec{\mathbf{F}}_3$?

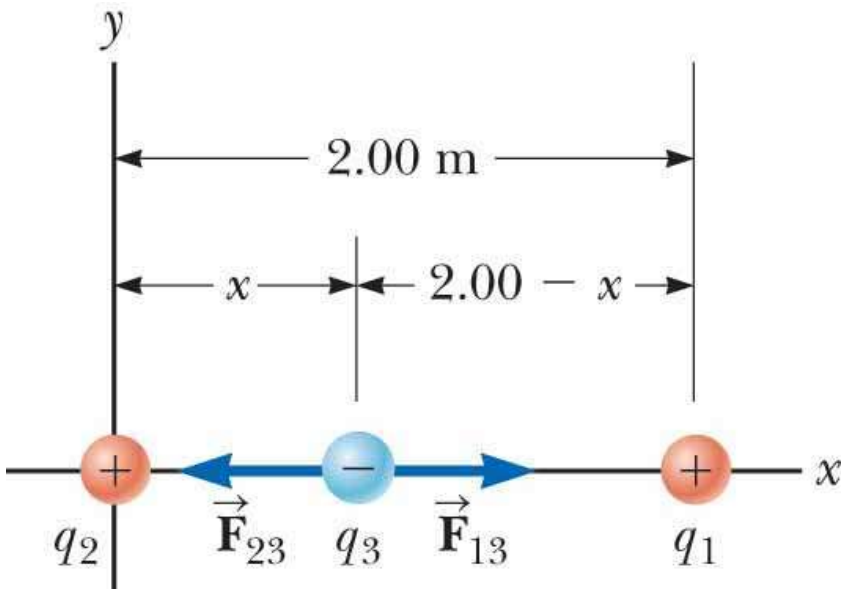


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Example 22.3: Where Is the Net Force Zero? (1 of 3)

Three point charges lie along the x axis as shown in the figure. The positive charge q_1 = 15.0 μC is at x = 2.00 m, the positive charge q_2 = 6.00 μC is at the origin, and the net force acting on q_3 is zero. What is the x coordinate of q_3?



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Example 22.3: Where Is the Net Force Zero? (2 of 3)

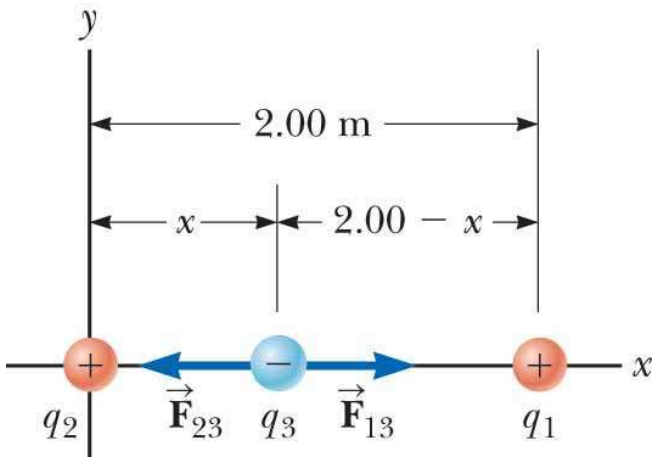
$$\sum \vec{\mathbf{F}}_3 = \vec{\mathbf{F}}_{23} + \vec{\mathbf{F}}_{13} = -k_e \frac{|q_2||q_3|}{x^2}\hat{\mathbf{i}} + k_e \frac{|q_1||q_3|}{(2.00-x)^2}\hat{\mathbf{i}} = 0$$

$$k_e \frac{|q_2||q_3|}{x^2} = k_e \frac{|q_1||q_3|}{(2.00-x)^2}$$

$$(2.00-x)^2 |q_2| = x^2 |q_1|$$

$$(2.00-x)\sqrt{|q_2|} = \pm x\sqrt{|q_1|}$$

$$x = \frac{2.00\sqrt{|q_2|}}{\sqrt{|q_2|} \pm \sqrt{|q_1|}}$$



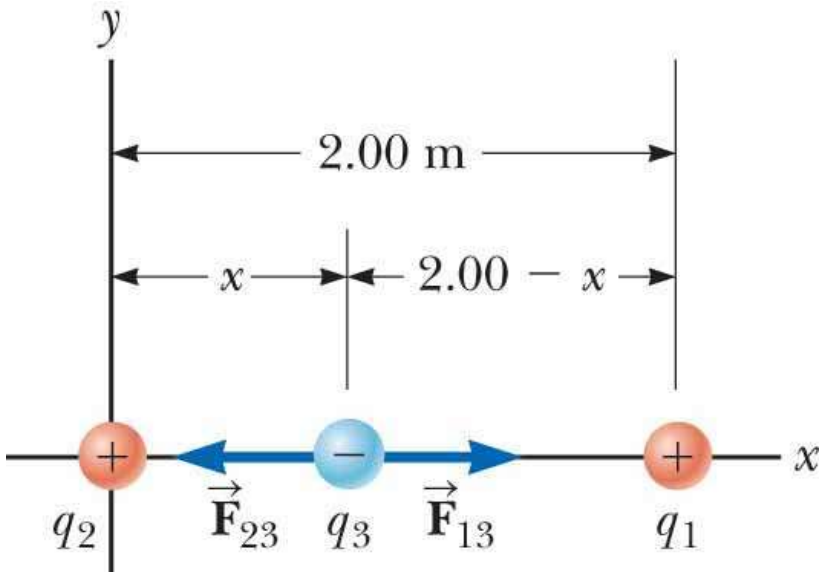
$$x = \frac{2.00\sqrt{6.00\times10^{-6}}}{\sqrt{6.00\times10^{-6}} + \sqrt{15.0\times10^{-6}}} = 0.775 \text{ m}$$

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Example 22.3: Where Is the Net Force Zero? (3 of 3)

Suppose q_3 is constrained to move only along the x axis. From its initial position at x = 0.775 m, it is pulled a small distance along the x axis. When released, does it return to equilibrium, or is it pulled farther from equilibrium? That is, is the equilibrium stable or unstable?

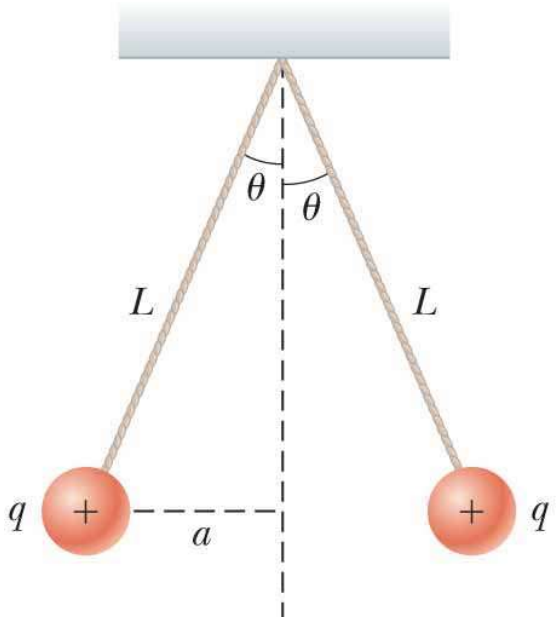


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Example 22.4: Find the Charge on the Spheres (1 of 3)

Two identical small charged spheres, each having a mass of 3.00×10^{-2} kg, hang in equilibrium as shown in the figure. The length L of each string is 0.150 m, and the angle θ is 5.00°. Find the magnitude of the charge on each sphere.



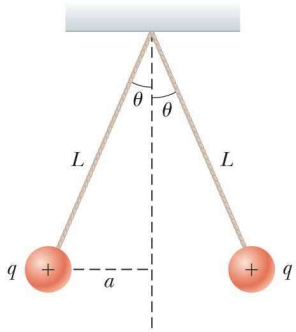
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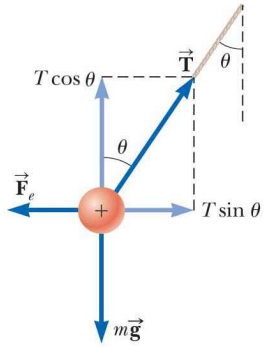
Example 22.4: Find the Charge on the Spheres (2 of 3)

$$\sum F_x = T\sin\theta - F_e = 0 \Rightarrow T\sin\theta = F_e$$
$$\sum F_y = T\cos\theta - mg = 0 \Rightarrow T\cos\theta = mg$$

$$\tan\theta = \frac{F_e}{mg} \Rightarrow F_e = mg\tan\theta$$



$$\sin\theta = \frac{a}{L} \Rightarrow a = L\sin\theta$$



$$|q| = \sqrt{\frac{F_e r^2}{k_e}} = \sqrt{\frac{F_e (2a)^2}{k_e}} = \sqrt{\frac{mg\tan\theta(2L\sin\theta)^2}{k_e}}$$

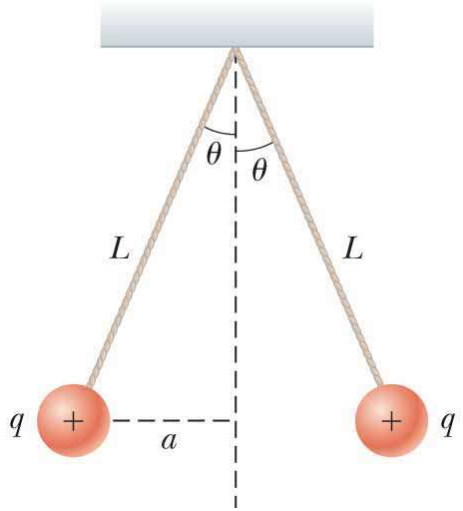
$$|q| = \sqrt{\frac{(3.00\times10^{-2}\ \text{kg})(9.80\ \text{m/s}^2)\tan(5.00°)[2(0.150\ \text{m})\sin(5.00°)]^2}{8.988\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2}} = 4.42\times10^{-8}\text{C}$$

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Example 22.4: Find the Charge on the Spheres (3 of 3)

Suppose your roommate proposes solving this problem without the assumption that the charges are of equal magnitude. She claims the symmetry of the problem is destroyed if the charges are not equal, so the strings would make two different angles with the vertical and the problem would be much more complicated. How would you respond?



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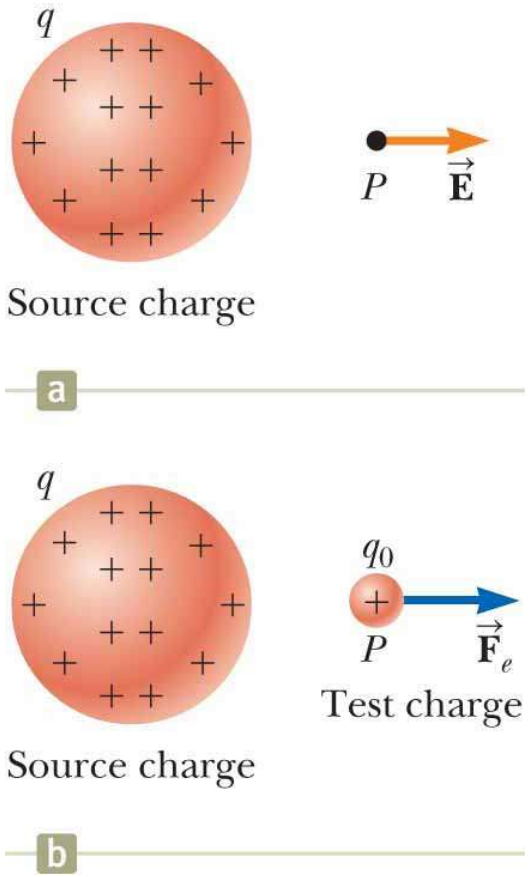
Analysis Model: Particle in a Field (Electric) (1 of 4)

$$\vec{\mathbf{g}} \equiv \frac{\vec{\mathbf{F}}_g}{m_0}$$

$$\vec{\mathbf{F}}_g = m\vec{\mathbf{g}}$$

$$\boxed{\vec{\mathbf{E}} \equiv \frac{\vec{\mathbf{F}}_e}{q_0}}$$

$$\boxed{\vec{\mathbf{F}}_e = q_0\vec{\mathbf{E}}}$$

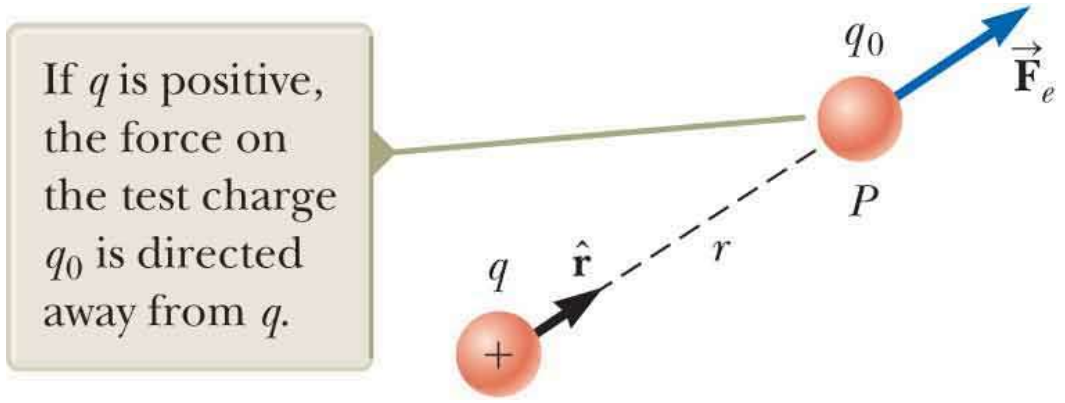


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Analysis Model: Particle in a Field (Electric) (2 of 4)

$$\vec{\mathbf{F}}_e = k_e \frac{qq_0}{r^2}\hat{\mathbf{r}} \qquad \vec{\mathbf{E}} = \vec{\mathbf{F}}_e / q_0 \qquad \boxed{\vec{\mathbf{E}} = k_e \frac{q}{r^2}\hat{\mathbf{r}}}$$



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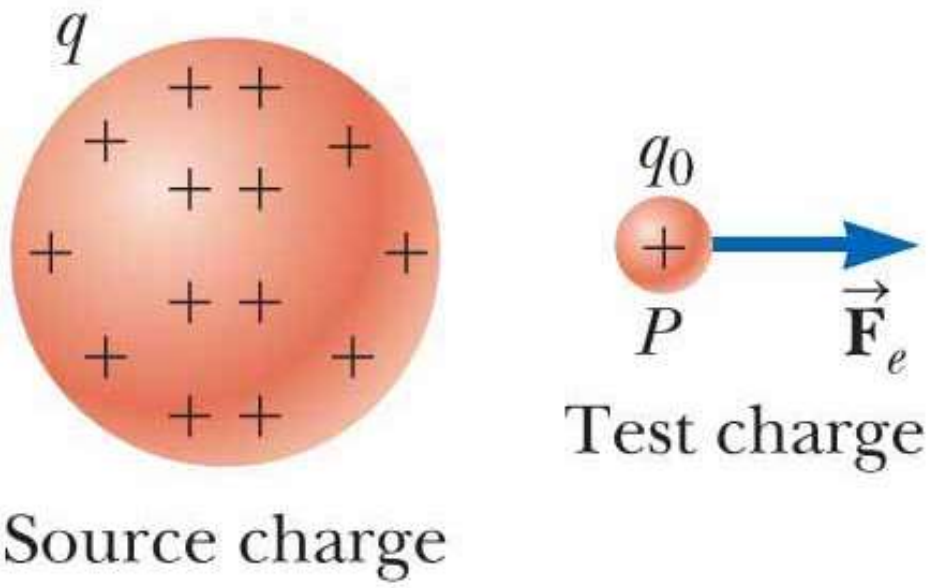
Analysis Model: Particle in a Field (Electric) (3 of 4)

$$\vec{\mathbf{g}} = \frac{\vec{\mathbf{F}}_g}{m_0} = -\frac{GM_E}{r^2}\hat{\mathbf{r}}$$

$$\vec{\mathbf{F}}_g = m\vec{\mathbf{g}}$$

$$\vec{\mathbf{E}} = k_e \frac{q}{r^2}\hat{\mathbf{r}}$$

$$\vec{\mathbf{F}}_e = q_0\vec{\mathbf{E}}$$



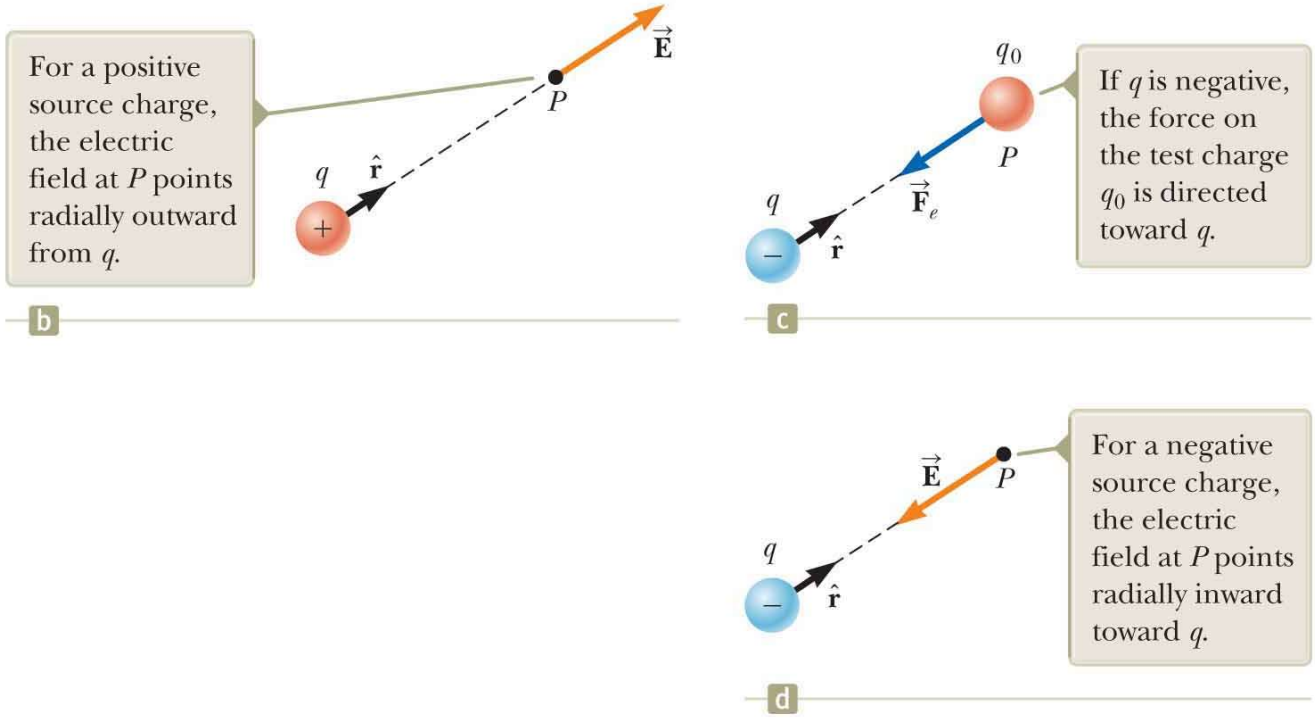
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Analysis Model: Particle in a Field (Electric) (4 of 4)

$$\vec{\mathbf{F}}_e = k_e \frac{qq_0}{r^2}\hat{\mathbf{r}}$$

$$\vec{\mathbf{E}} = k_e \sum_i \frac{q_i}{r_i^2}\hat{\mathbf{r}}_i$$



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Electric Force



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Quick Quiz 22.4 (1 of 2)

A test charge of +3 μC is at a point *P* where an external electric field is directed to the right and has a magnitude of 4×10^6 N/C. If the test charge is replaced with another test charge of –3 μC, what happens to the external electric field at *P*?

(a) It is unaffected.

(b) It reverses direction.

(c) It changes in a way that cannot be determined.

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Quick Quiz 22.4 (2 of 2)

A test charge of +3 μC is at a point *P* where an external electric field is directed to the right and has a magnitude of 4×10^6 N/C. If the test charge is replaced with another test charge of –3 μC, what happens to the external electric field at *P*?

(a) It is unaffected.

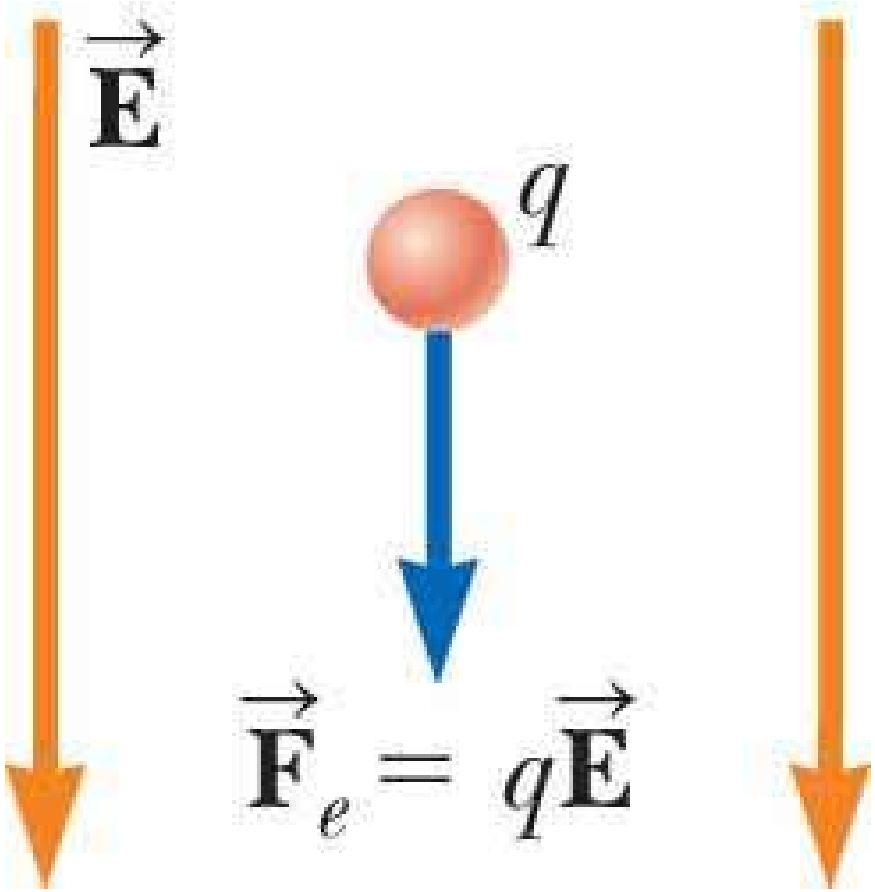
(b) It reverses direction.

(c) It changes in a way that cannot be determined.

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Analysis Model: Particle in a Field (Electric)



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Example 22.5: A Suspended Water Droplet

A water droplet of mass 3.00×10^{-12} kg is located in the air near the ground during a stormy day. An atmospheric electric field of magnitude 6.00×10^{3} N/C points vertically downward in the vicinity of the water droplet. The droplet remains suspended at rest in the air. What is the electric charge on the droplet?

$$\sum F_y = 0 \rightarrow F_e - F_g = 0 \qquad \Rightarrow q(-E) - mg = 0$$

$$q = -mg/E$$

$$q = -\frac{\left(3.00\times10^{-12}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)}{6.00\times10^{3}\ \text{N/C}} = \boxed{-4.90\times10^{-15}\text{C}}$$

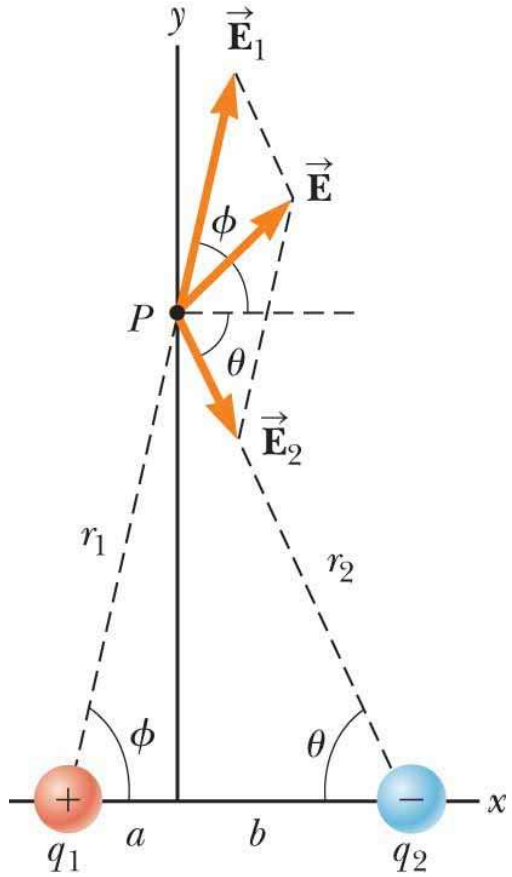
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Example 22.6: Electric Field Due to Two Charges (1 of 4)

Charges q_1 and q_2 are located on the x axis, at distances a and b, respectively, from the origin as shown in the figure.

(A) Find the components of the net electric field at the point P, which is at position (0, y).



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Example 22.6: Electric Field Due to Two Charges (2 of 4)

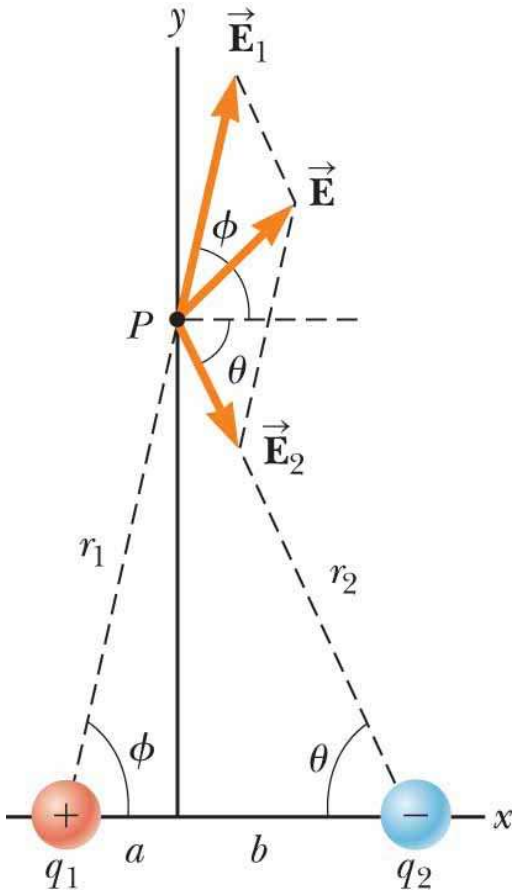
$$E_1 = k_e \frac{|q_1|}{r_1^2} = k_e \frac{|q_1|}{a^2 + y^2} \qquad E_2 = k_e \frac{|q_2|}{r_2^2} = k_e \frac{|q_2|}{b^2 + y^2}$$

$$\vec{\mathbf{E}}_1 = k_e \frac{|q_1|}{a^2 + y^2} \cos\phi\,\hat{\mathbf{i}} + k_e \frac{|q_1|}{a^2 + y^2} \sin\phi\,\hat{\mathbf{j}}$$

$$\vec{\mathbf{E}}_2 = k_e \frac{|q_2|}{b^2 + y^2} \cos\theta\,\hat{\mathbf{i}} - k_e \frac{|q_2|}{b^2 + y^2} \sin\theta\,\hat{\mathbf{j}}$$

$$E_x = k_e \frac{|q_1|}{a^2 + y^2} \cos\phi + k_e \frac{|q_2|}{b^2 + y^2} \cos\theta$$

$$E_y = k_e \frac{|q_1|}{a^2 + y^2} \sin\phi - k_e \frac{|q_2|}{b^2 + y^2} \sin\theta$$



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Example 22.6: Electric Field Due to Two Charges (3 of 4)

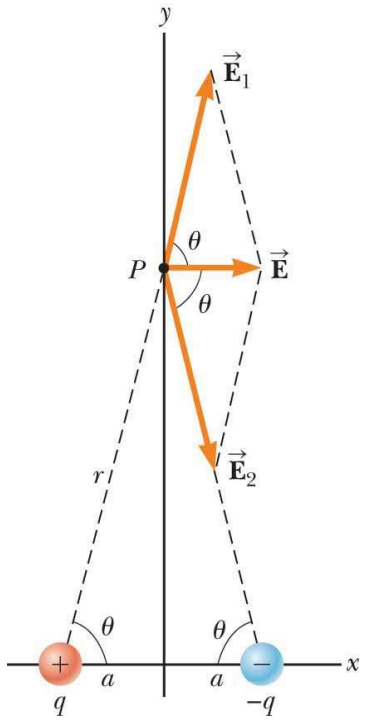
(B) Evaluate the electric field at point *P* in the special case that $|q_1| = |q_2|$ and $a = b$.

$$E_x = k_e \frac{q}{a^2 + y^2}\cos\theta + k_e \frac{q}{a^2 + y^2}\cos\theta = 2k_e \frac{q}{a^2 + y^2}\cos\theta$$

$$E_y = k_e \frac{q}{a^2 + y^2}\sin\theta - k_e \frac{q}{a^2 + y^2}\sin\theta = 0$$

$$\cos\theta = \frac{a}{r} = \frac{a}{\left(a^2 + y^2\right)^{1/2}}$$

$$E_x = 2k_e \frac{q}{a^2 + y^2}\left[\frac{a}{\left(a^2 + y^2\right)^{1/2}}\right] = \boxed{k_e \frac{2aq}{\left(a^2 + y^2\right)^{3/2}}}$$



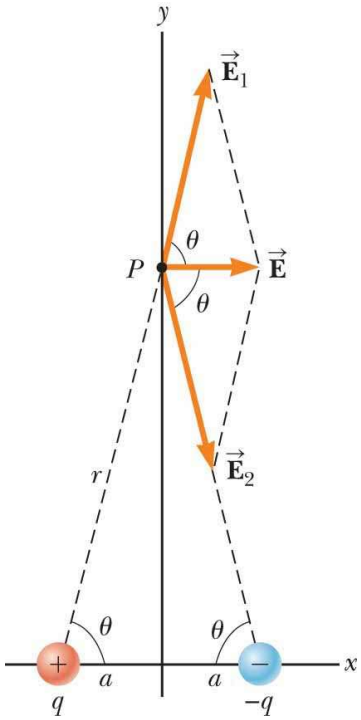
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Example 22.6: Electric Field Due to Two Charges (4 of 4)

(C) Find the electric field due to the electric dipole when point *P* is a distance *y* >> *a* from the origin.

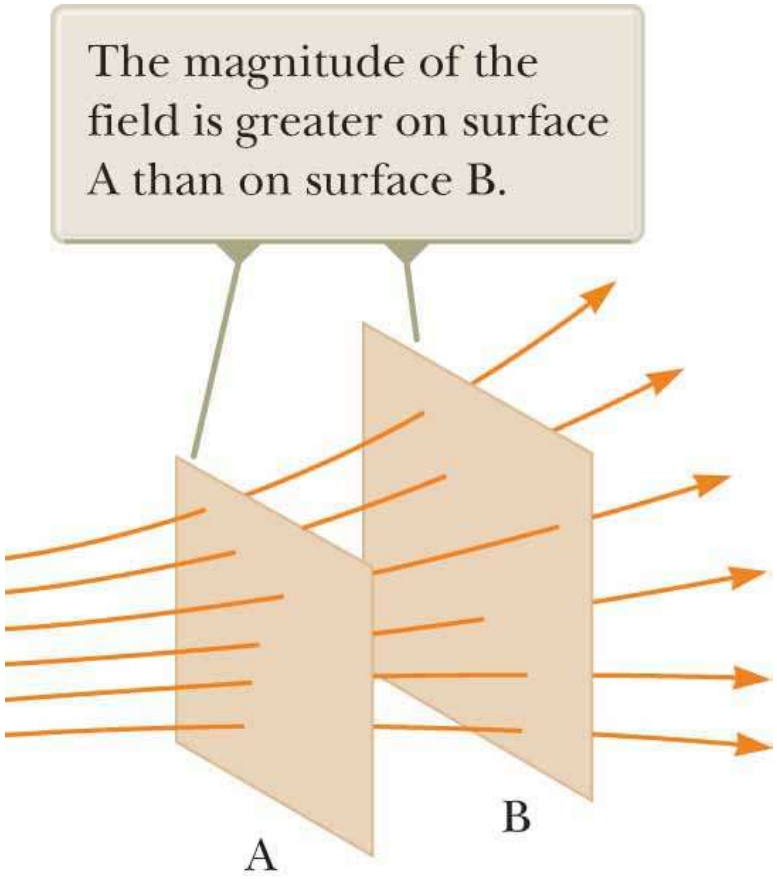
$$E_x = k_e \frac{2aq}{\left(a^2 + y^2\right)^{3/2}} \Rightarrow E \approx k_e \frac{2aq}{y^3}$$



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Electric Field Lines (1 of 5)

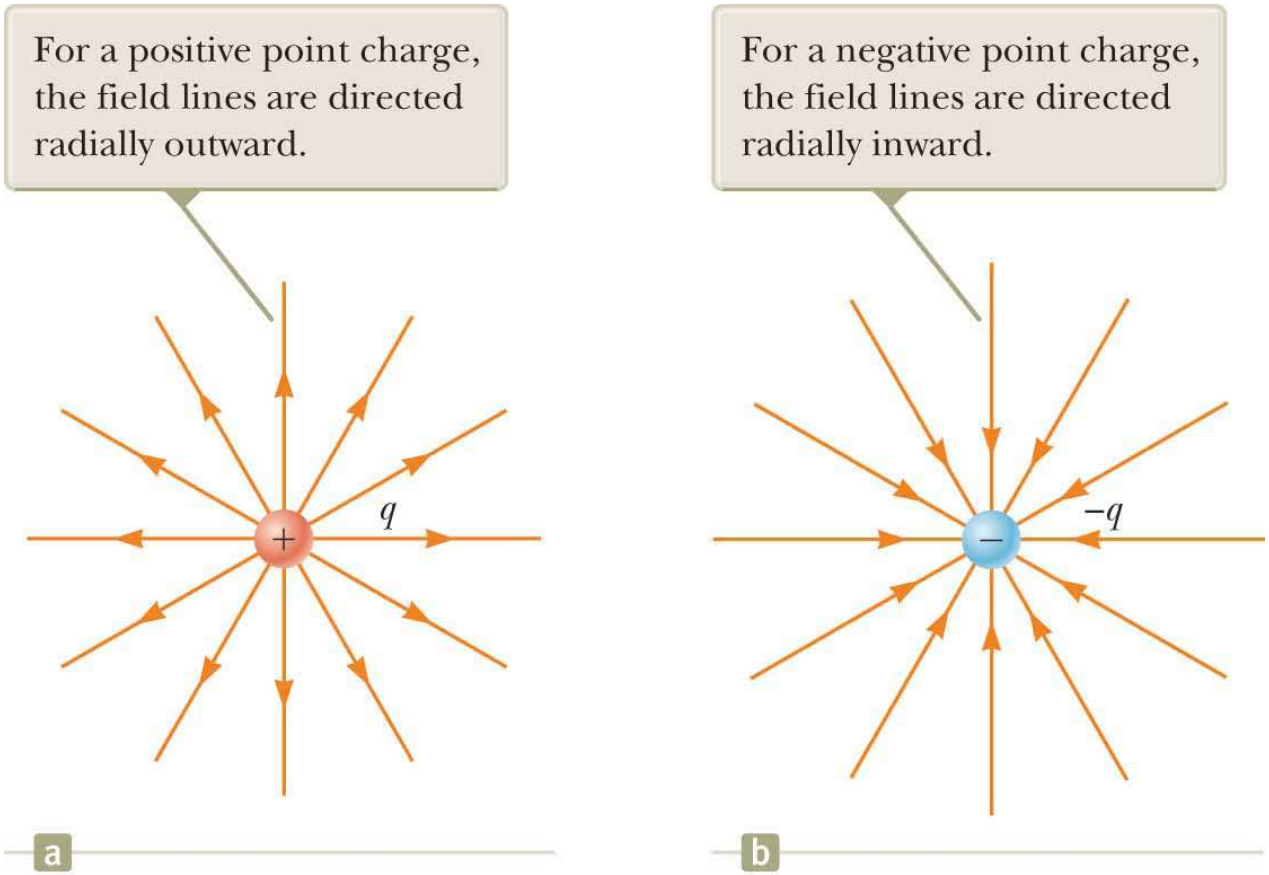


$$\vec{\mathbf{E}} = k_e \sum_i \frac{q}{r^2} \hat{\mathbf{r}}$$

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Electric Field Lines (2 of 5)

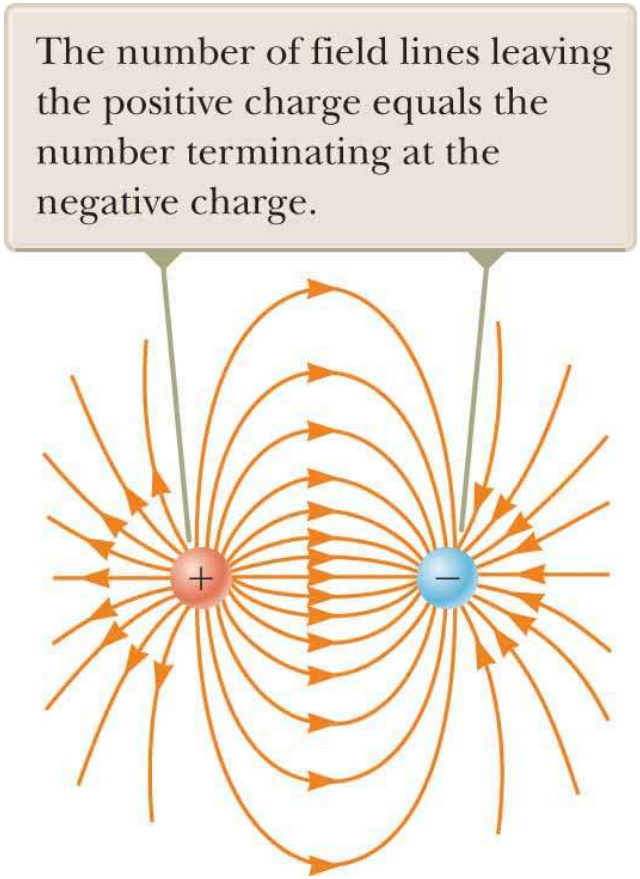


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Electric Field Lines (3 of 5)

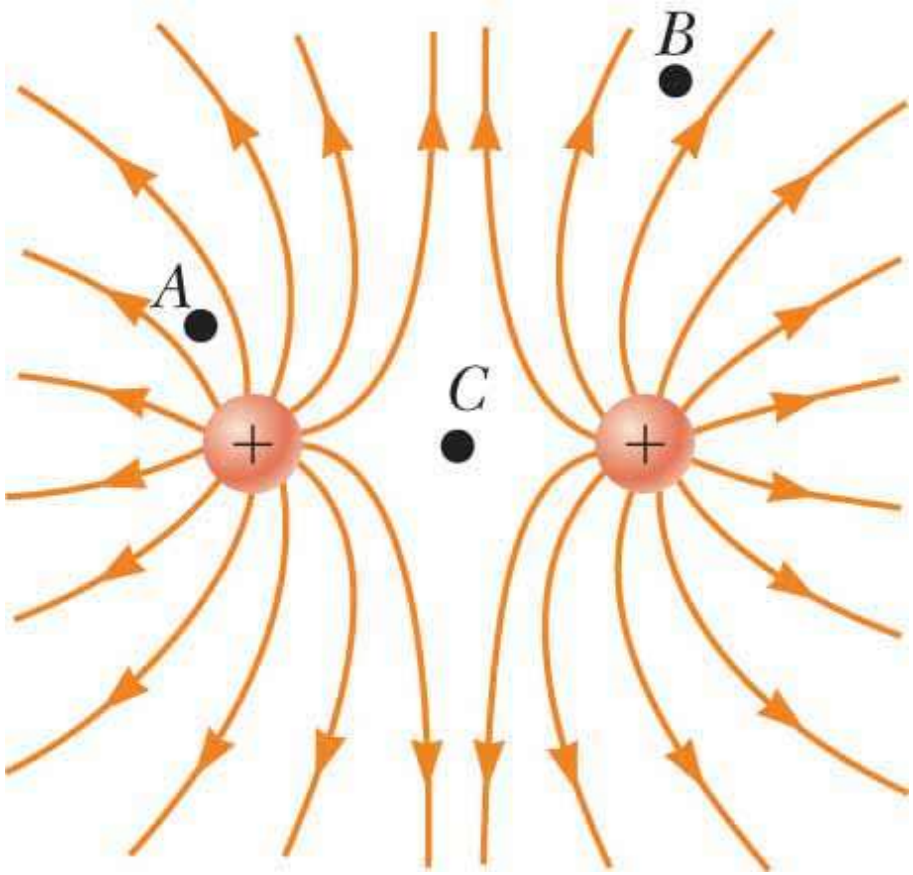
$$\frac{N_2}{N_1} = \left|\frac{Q_2}{Q_1}\right|$$



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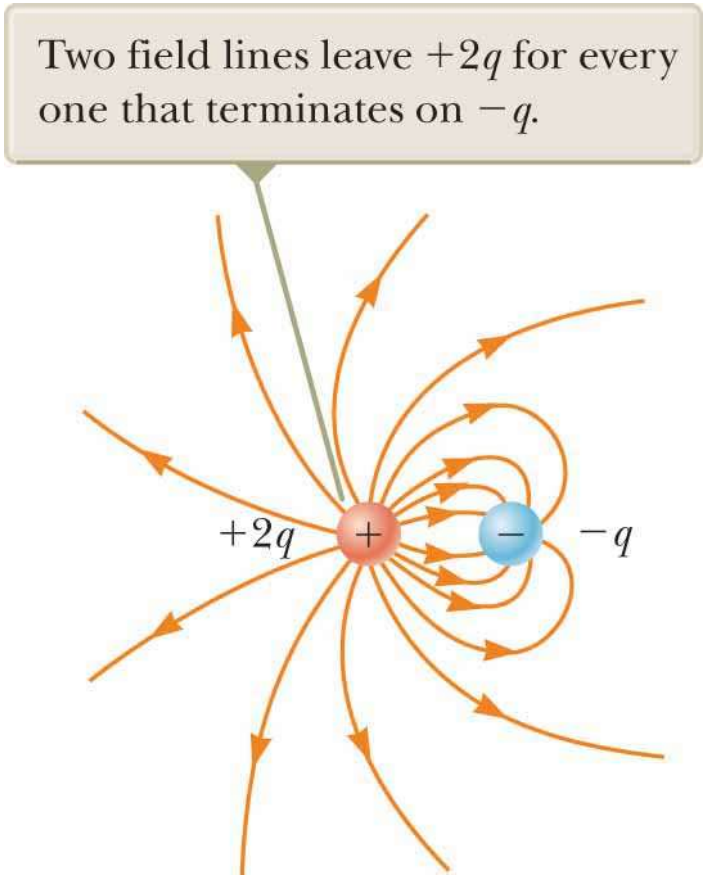
Electric Field Lines (4 of 5)



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Electric Field Lines (5 of 5)

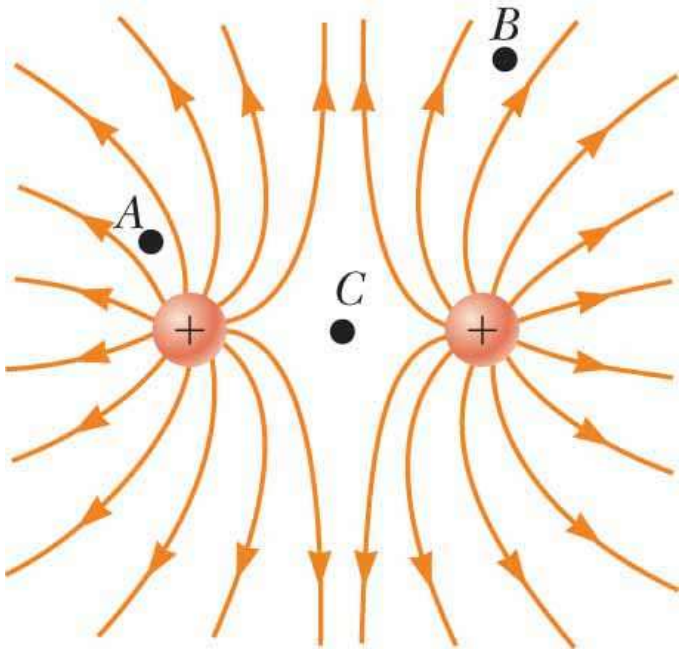


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Quick Quiz 22.5 (1 of 2)

Rank the magnitudes of the electric field at points A, B, and C shown in the figure (greatest magnitude first).



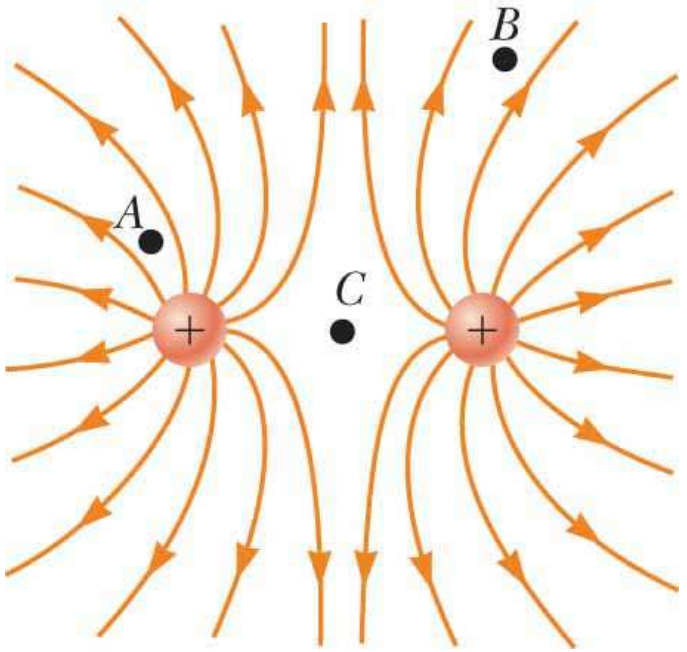
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Quick Quiz 22.5 (2 of 2)

Rank the magnitudes of the electric field at points A, B, and C shown in the figure (greatest magnitude first).

A, B, C



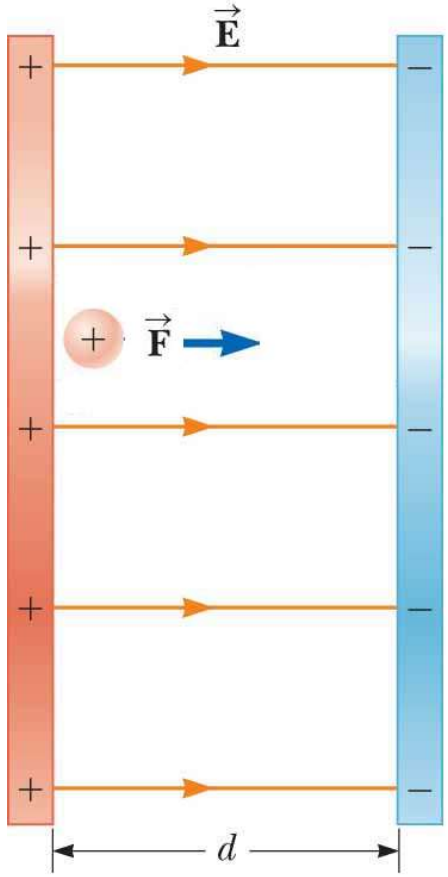
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Motion of a Charged Particle in a Uniform Electric Field

$$\sum \vec{\mathbf{F}} = q\vec{\mathbf{E}} = m\vec{\mathbf{a}}$$

$$\vec{\mathbf{a}} = \frac{q\vec{\mathbf{E}}}{m}$$



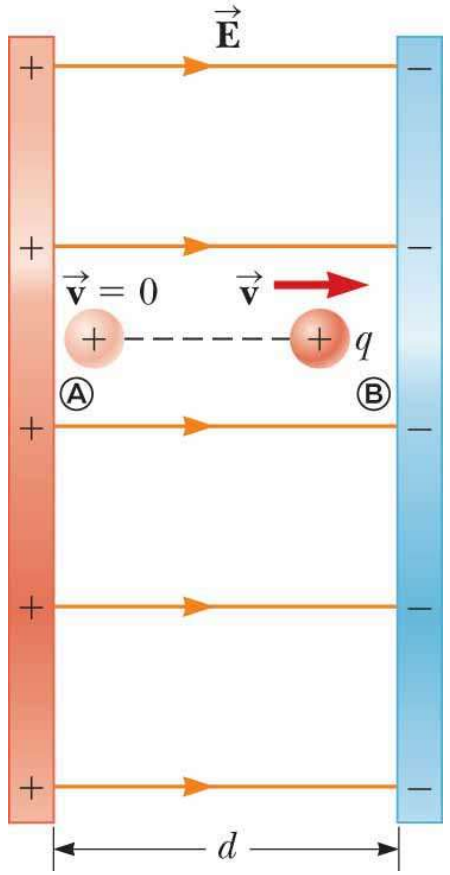
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Example 22.7: An Accelerating Positive Charge: Two Models (1 of 3)

A uniform electric field $\vec{\mathbf{E}}$ is directed along the *x* axis between parallel plates of charge separated by a distance *d* as shown in the figure. A positive point charge *q* of mass *m* is released from rest at a point A next to the positive plate and accelerates to a point B next to the negative plate.

(A) Find the speed of the particle at B by modeling it as a particle under constant acceleration.



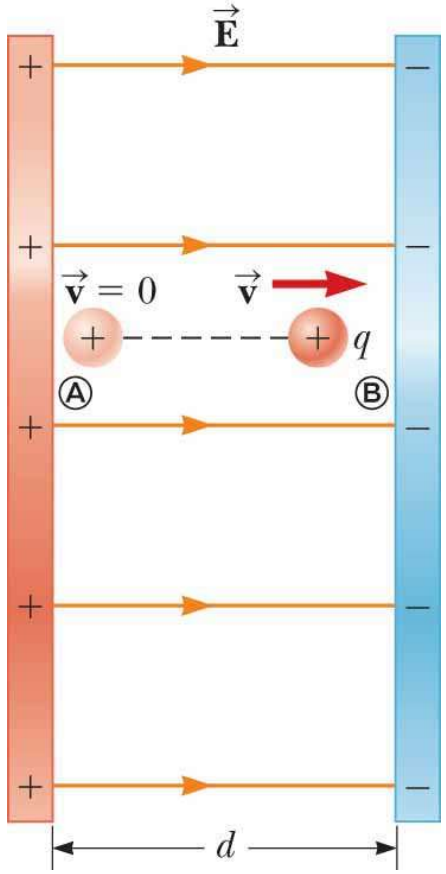
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Example 22.7: An Accelerating Positive Charge: Two Models (2 of 3)

$$v_f^2 = v_i^2 + 2a\left(x_f - x_i\right) = 0 + 2a\left(d - 0\right) = 2ad$$

$$v_f = \sqrt{2ad} = \sqrt{2\left(\frac{qE}{m}\right)d} = \boxed{\sqrt{\frac{2qEd}{m}}}$$



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Example 22.7: An Accelerating Positive Charge: Two Models (3 of 3)

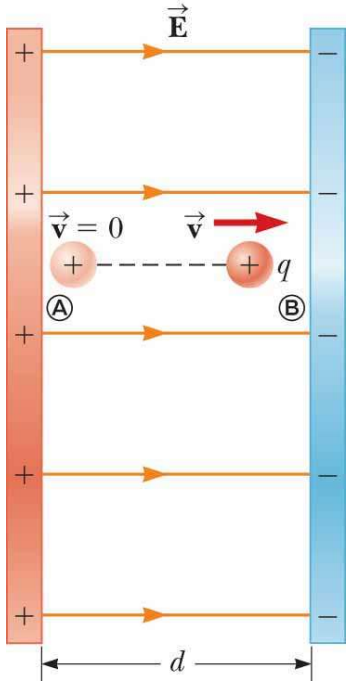
(B) Find the speed of the particle at B by modeling it as a nonisolated system in terms of energy.

$$W = \Delta K$$

$$F_e \Delta x = K_B - K_A = \frac{1}{2} m v_f^{\,2} - 0$$

$$\Rightarrow v_f = \sqrt{\frac{2F_e \Delta x}{m}}$$

$$v_f = \sqrt{\frac{2(qE)(d)}{m}} = \boxed{\sqrt{\frac{2qEd}{m}}}$$



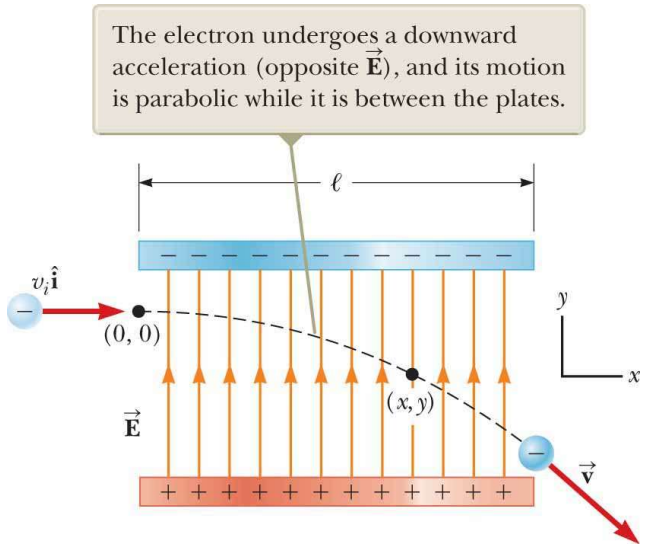
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Example 22.8: An Accelerated Electron (1 of 4)

An electron enters the region of a uniform electric field as shown in the figure, with $v_i = 3.00 \times 10^6$ m/s and E = 200 N/C. The horizontal length of the plates is $\ell = 0.100$ m.

(A) Find the acceleration of the electron while it is in the electric field.



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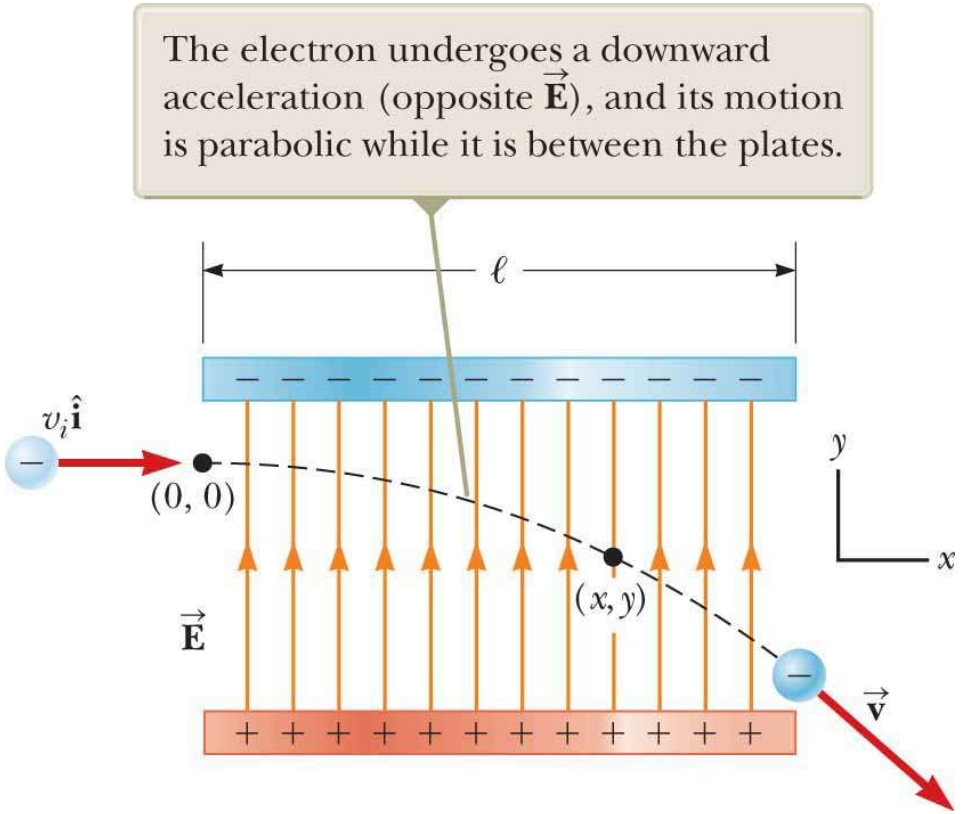
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Example 22.8: An Accelerated Electron (2 of 4)

$$a_y = -\frac{eE}{m_e}$$

$$a_y = -\frac{\left(1.60\times10^{-19}\ \text{C}\right)\left(200\ \text{N/C}\right)}{9.11\times10^{-31}\ \text{kg}}$$

$$= \boxed{-3.51\times10^{13}\ \text{m/s}^2}$$



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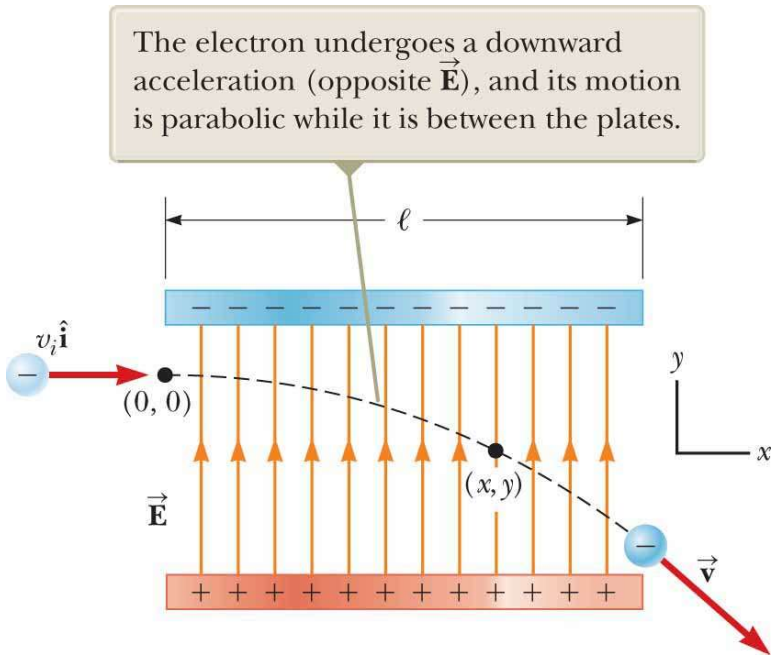
Example 22.8: An Accelerated Electron (3 of 4)

(B) Assuming the electron enters the field at time $t = 0$, find the time at which it leaves the field.

$$x_f = x_i + v_x t \Rightarrow t = \frac{x_f - x_i}{v_x}$$

$$t = \frac{\ell - 0}{v_x} = \frac{0.100 \text{ m}}{3.00 \times 10^6 \text{ m/s}}$$

$$= \boxed{3.33 \times 10^{-8} \text{ s}}$$

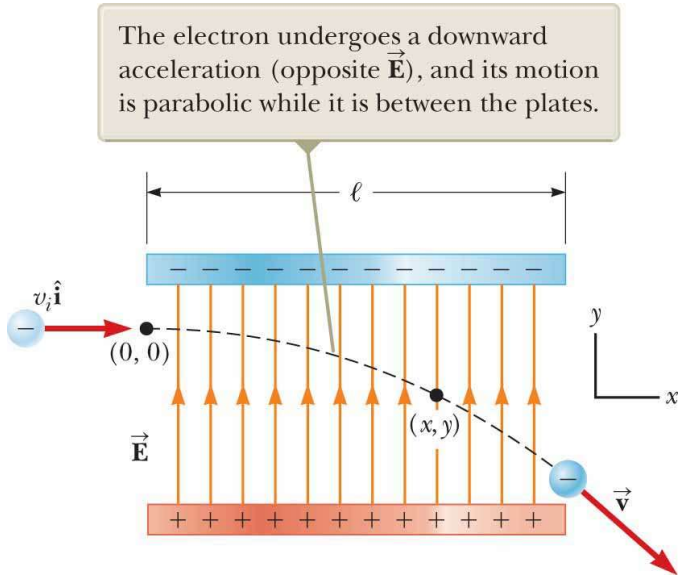


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Example 22.8: An Accelerated Electron (4 of 4)

(C) Assuming the vertical position of the electron as it enters the field is $y_i = 0$, what is its vertical position when it leaves the field?



$$y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

$$y_f = 0 + 0 + \frac{1}{2}\left(-3.51\times10^{13}\ \text{m/s}^2\right)\left(3.33\times10^{-8}\ \text{s}\right)^2$$

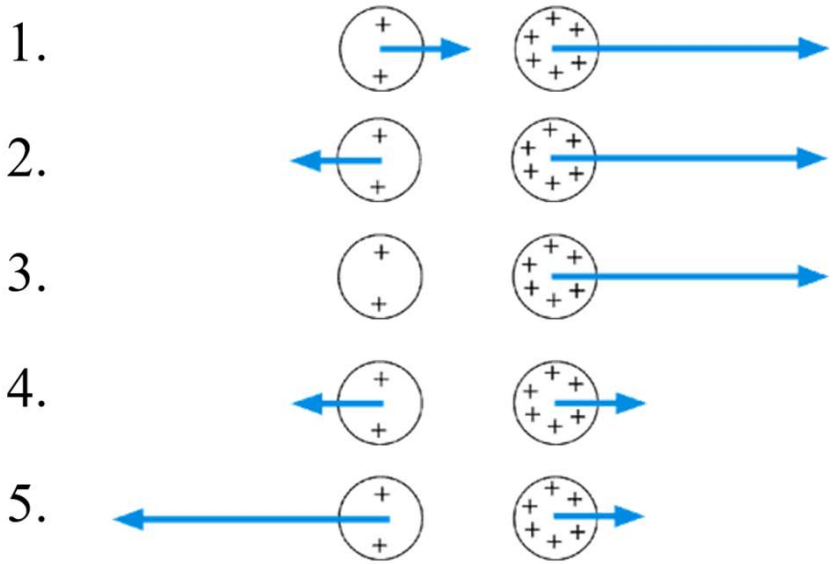
$$= -0.0195\ \text{m} = \boxed{-1.95\ \text{cm}}$$

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Assessing to Learn (1 of 3)

The diagrams below show two uniformly charged spheres. The charge on the right sphere is 3 times as large as the charge on the left sphere. Which force diagram best represents the magnitudes and directions of the electric forces on the two spheres?

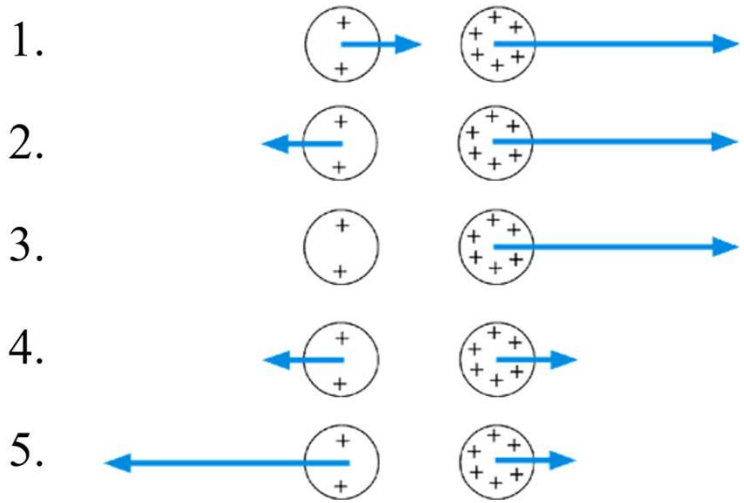


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Assessing to Learn (2 of 3)

The diagrams below show two uniformly charged spheres. The charge on the right sphere is 3 times as large as the charge on the left sphere. Each arrow represents the electric field at the center of one sphere created by the other. Which choice best represents the magnitudes and directions of the electric field vectors created by one sphere at the location of the other sphere?



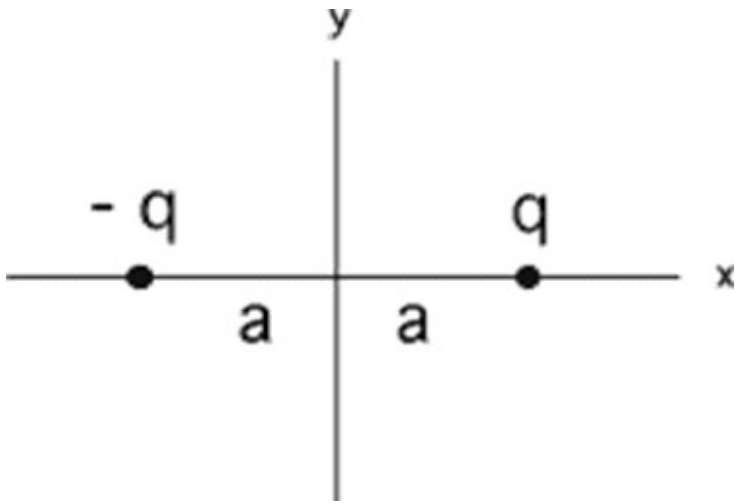
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Assessing to Learn (3 of 3)

Where, other than at infinity, is the electric field 0 in the vicinity of the dipole shown?

1. Along the *y*-axis.
2. At the origin.
3. At two points, one to the right of (a, 0), the other to the left of (−a, 0).
4. At two points on the *y*-axis, one below the origin, one above the origin.
5. None of the above.



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